Non-invasive AI Agent for AI Phone

FluidGPT

The Winner of 1st AI Championship

AI CHAMPION

KAIST

fluiz

01

Introduction

FluidGPT

1 INTRODUCTION | Pain Point

Smartphones have evolved, but how we use them hasn’t

A smartphone we use all day

Problems of Touch-Based Smartphone Use

Safety / Productivity

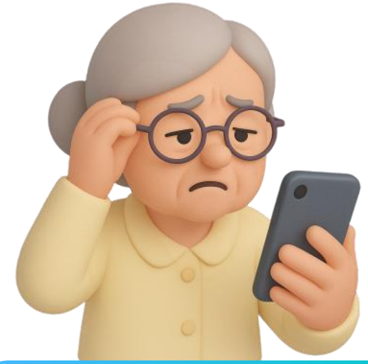
App navigation

▶ Time-comsuming & inefficient

Cognitive Overload

Complex interfaces

▶ Cognitive effort & fatigue



Digital barriers

Complexity
For vulnerable users

▶ Access barrierrs

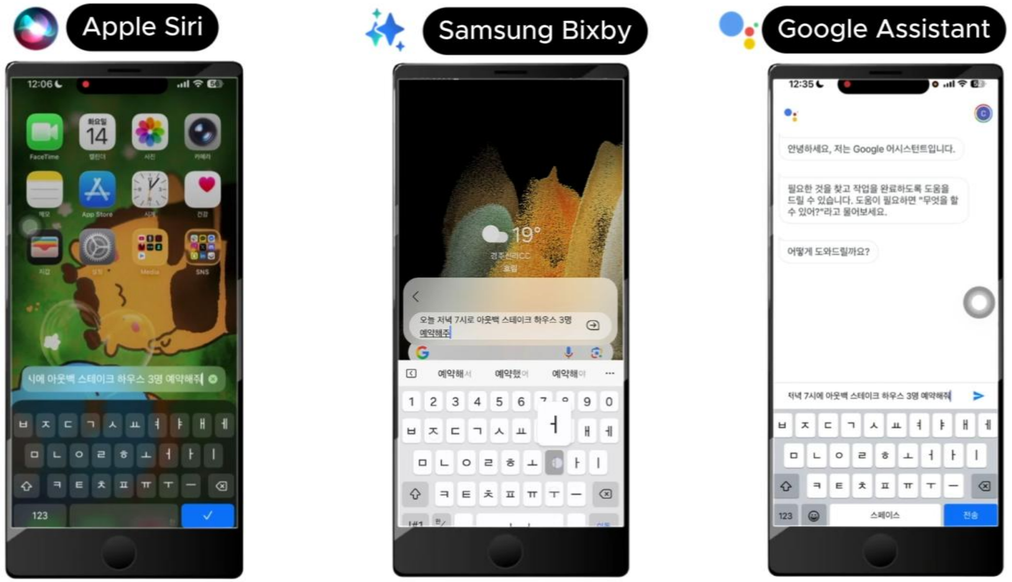
In the AI era, why are we still tapping apps?

High Failure Rates in Autonomous App Execution for Current Voice Assistants

Failure Rate

Booking a Restaurant with Mobile Agents

P



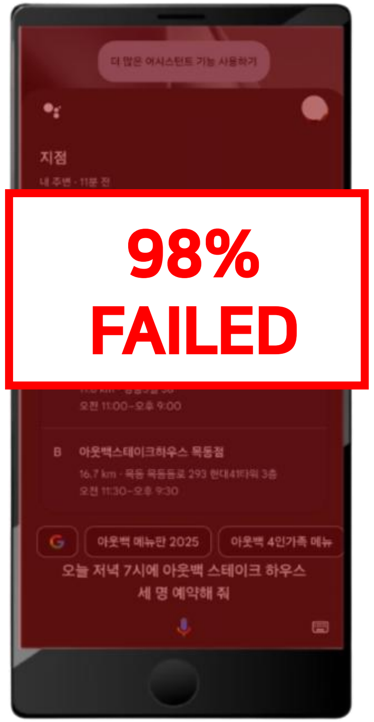
Test Results

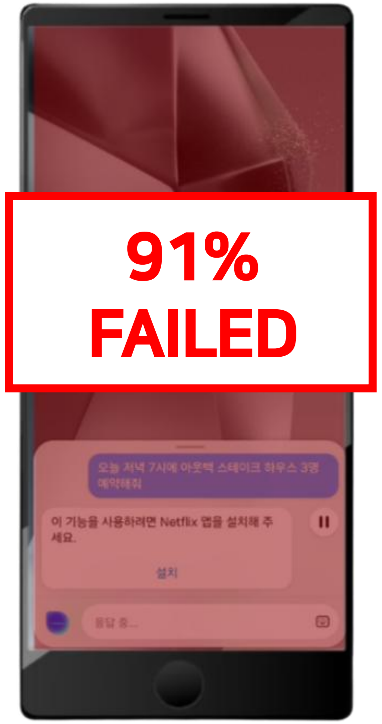
Top 100 Apps	Apple Siri	Google Assistant	Samsung Bixby
	X	X	X
	O	O	O
	X	X	X
	X	X	X
	X	X	X
	X	X	X
	X	X	X
	X	X	X
⋮			
	X	X	X
	99% Failure	98% Failure	91% Failure

High Failure Rates in Autonomous App Execution for Current AI Agents

Failure Rate

Apple Siri | **Google Assistant** | **Samsung Bixby**





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⋮			
	X	X	X
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Introducing FluidGPT, a **Mobile Agent** that executes apps by voice

A generative-AI agent that autonomously executes existing apps in a **non-invasive** and **general-purpose** way.

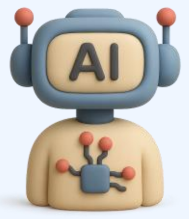
No APIs required,
No code modifications needed

Generative AI NLU

Just **Speak**

→

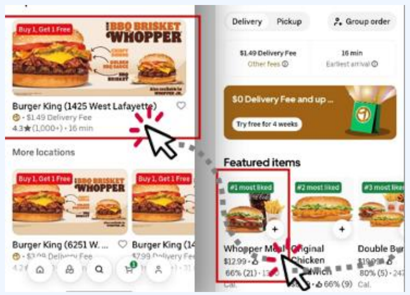
FluidGPT



Runs existing apps/web **automatically**

→

Non-invasive Excution



Executes user commands **autonomously**

Widespread Use

Life

Taxi request

Healthcare

Hospital appointment

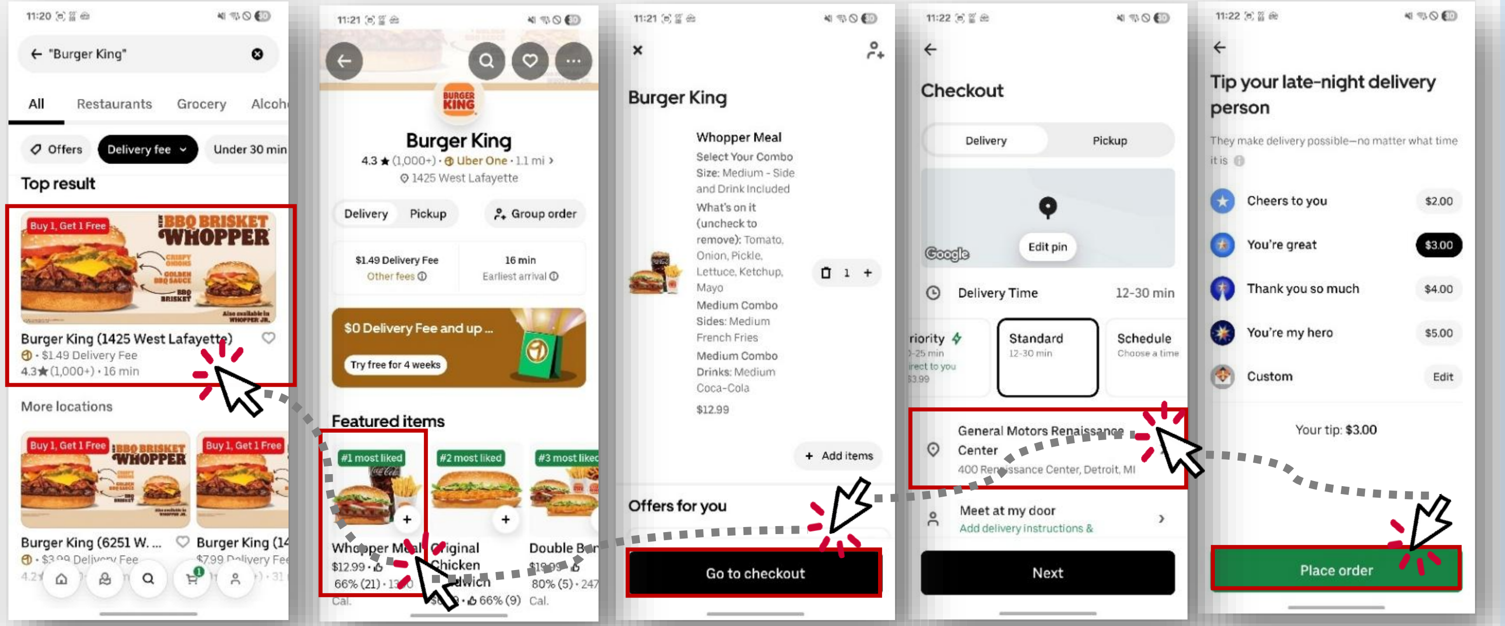
Shopping

Grocery shopping

| FluidGPT Demo : Order from the UberEats |

How FluidGPT launches UberEats and executes the user's instruction

Conversation between the user and an autonomous app-execution agent



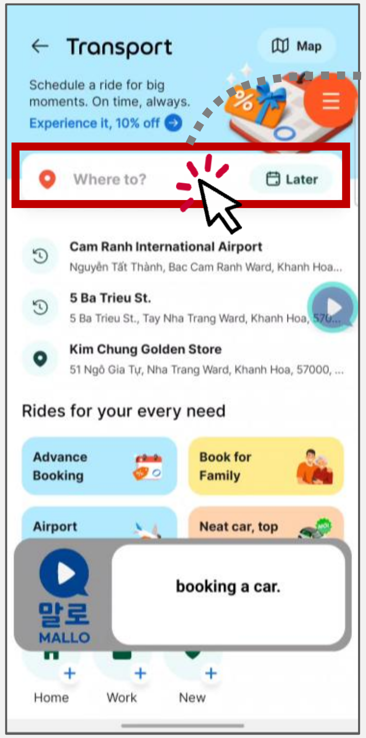
Search "Burger King" | Click "Whopper Meal" | Click "Checkout" | Select "Address" | Click "Order"

Demo Clip

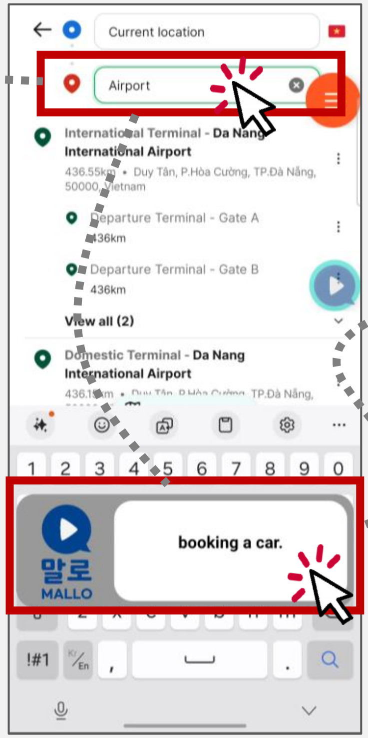
| FluidGPT Demo : Calling a Grab ride |

How FluidGPT launches Grab and executes the user's instructions

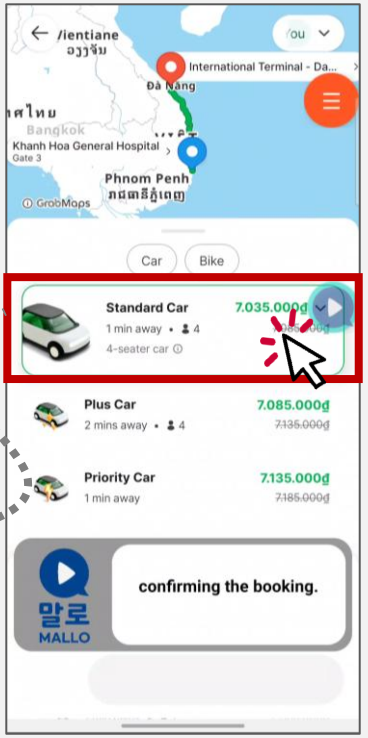
Conversation between the user and an autonomous app-execution agent



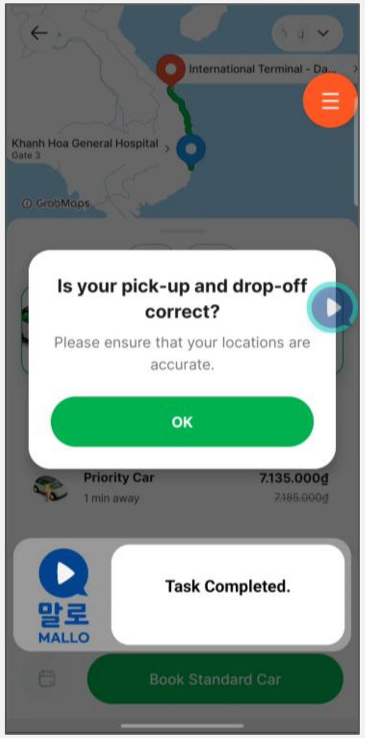
Click the destination



Enter the destination

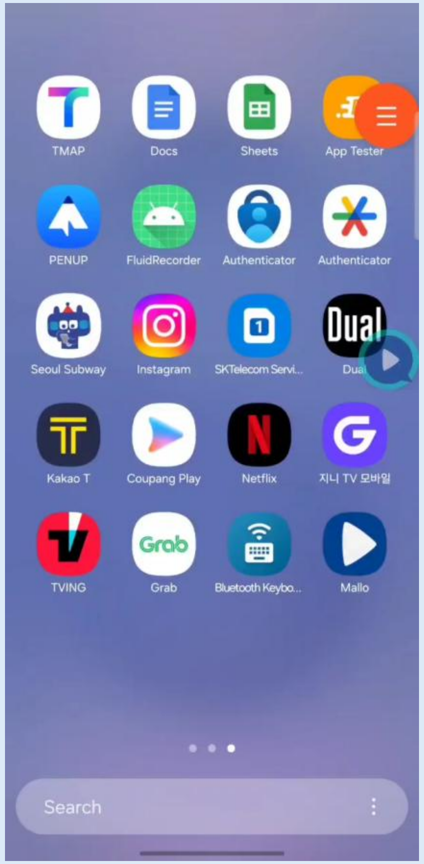


Select the taxi type



Task Completed

▶ Demo Clip



| FluidGPT Demo : Calling a Kakao Taxi |

FluidGPT

Call a taxi

FluidGPT

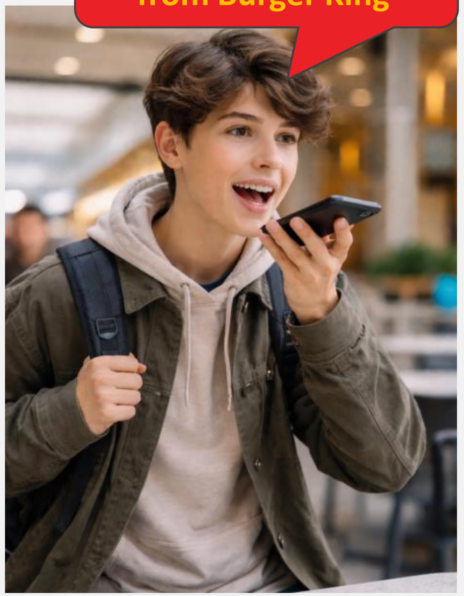
Call a taxi
to the airport.

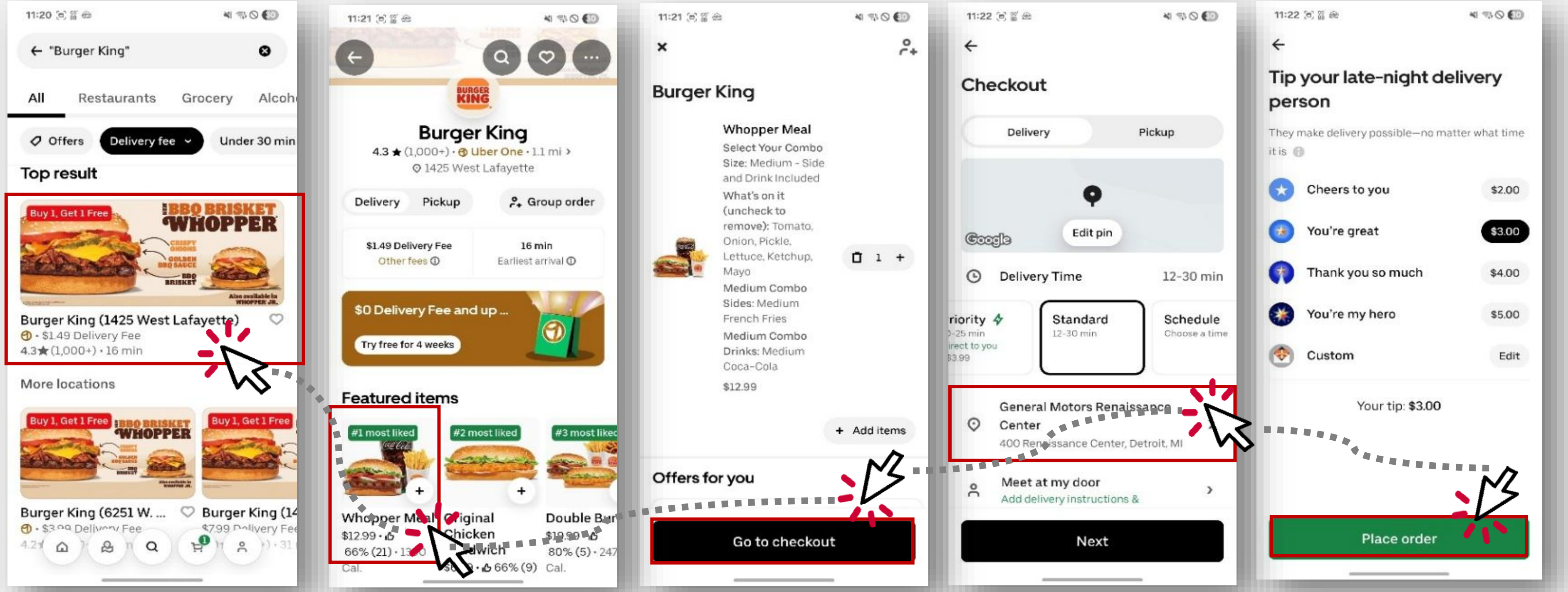
| FluidGPT Demo : Order from the UberEats |

How FluidGPT launches UberEats and executes the user's instruction

Conversation between the user and an autonomous app-execution agent

Can I Get some Burger from Burger King





Search "Burger King"

Click "Whopper Meal"

Click "Checkout"

Select "Address"

Click "Order"

02

Technology Overview

FluidGPT

| Technical Limitations of Existing LLM-Based Mobile Agents |

Low Reliability

Accuracy at 70–80% based on SoTA models

High Inference Cost

Problem 1
Lack of reproducibility

LLMs may choose different clicks or paths even under identical screen and input conditions

▶ Decreased consistency in execution results

Problem 2
Vulnerability in Multi-step Tasks

Reasoning accuracy drops sharply as context length increases

▶ Higher risk of error accumulation in multi-step tasks

Problem 3
UI Diversity and Non-standardization

UI structures vary by device and version, and non-standard UIs exist

▶ Difficult to accurately recognize and operate

Problem 4
Latency and cost issues

App control involves multi-step interactions (tap, scroll, input), requiring LLM calls at every step

▶ Constraints on economic viability and real-time responsiveness

| Neuro-Symbolic Hybrid Execution |

Hybrid app execution : neural flexibility + symbolic robustness

Stores diverse information required for app execution
(e.g., app info, task guidelines)

Stores various app sub-tasks in algorithmic form
(e.g., search product, add to cart, enter address)

Handles unforeseen situations occurring during program execution
(e.g., pop-ups, loading errors, insufficient balance)

Performs intelligent tasks requiring reasoning
(e.g., execution planning, product comparison, Q&A)

Vertical AI specialized for autonomous app execution

FluidGPT Architecture

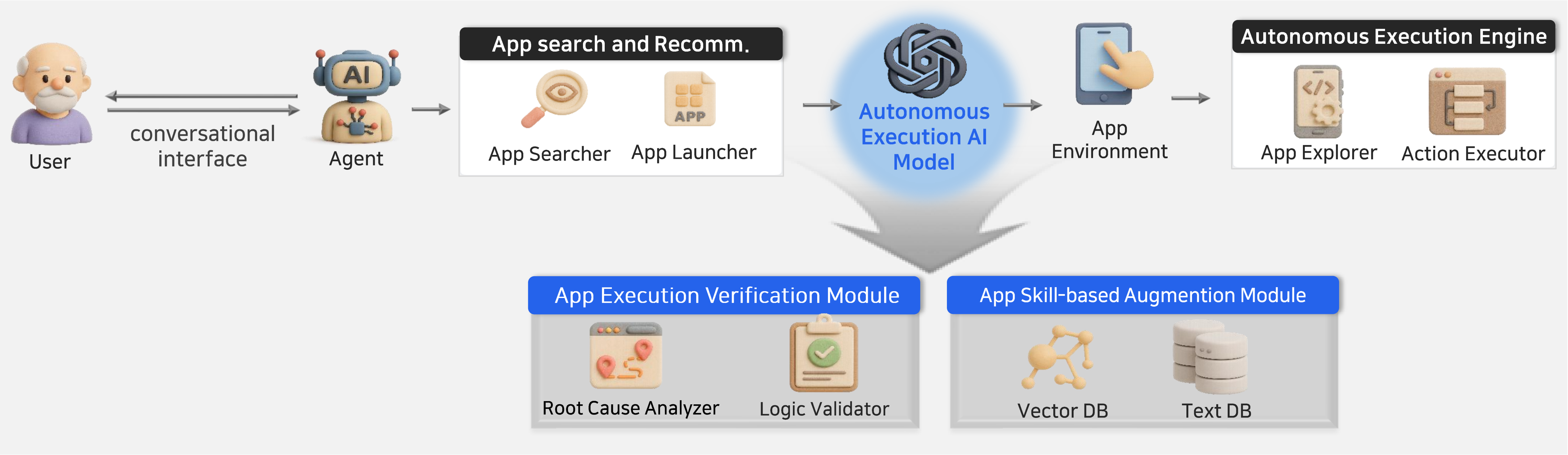
Execution Pipeline | Knowledge Augmentation

Core Modules Map :

① NL Interface ② App Search ③ Auto Model
④ Know. Base ⑤ Verification ⑥ Exec. Engine

World's 1st
Adopted logic-based formal Verification methods

Step 1 (Core Module ①)	Step 2 (Core Module ②)	Step 3 (Core Modules ③,⑥)	Step 4 (Core Module ⑤)
Conversation	**App Search**	**Autonomous Exec.**	**Verification**
NL Interface	Recommendation	Engine Core	Safety Check



| FluidGPT Core Features |

Perception (Visual & Structural)

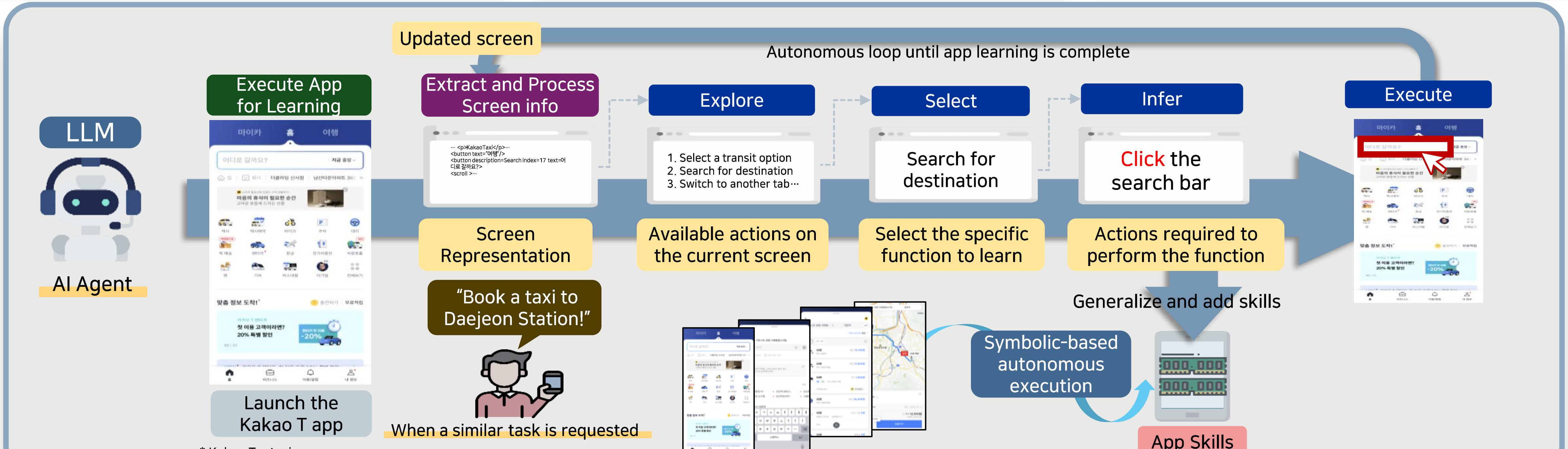
- UI differs by app & non-standard structures → consistent recognition is difficult
- Recognizes screen UI visually and structurally & normalizes by functional units (Perception-based Schema) → Enables consistent recognition and universal learning across apps

App Skills Construction (Skill Learning)

- Unclear boundaries between app functions & varying behavioral patterns across apps
- Learns app behavior patterns and extracts Skills through an "Explore-Select-Infer-Execute" process → ensures function-level generalization and reusability (App-Intent-based abstraction)

App Skills Utilization (Skill Execution)

- Difficulty in context-aware Skill utilization and optimal action determination
- Adaptive Inference Engine analyzes execution context to select Skills accordingly → intelligent·flexible handling across diverse scenarios



* Kakao T : taxi app

| FluidGPT: Patents & Awards Track Record |

ACM MobiCom Papers
2024. 11


MobileGPT: Augmenting LLM with Human-like App Memory for Mobile Task Automation

Sunjae Lee[1,*], Junyoung Choi[1], Jungjae Lee[1], Munim Hasan Wasi[1], Hojun Choi[1], Steven Y. Ko[2], Sangeun Oh[3,†], Insik Shin[1,4,*,†]

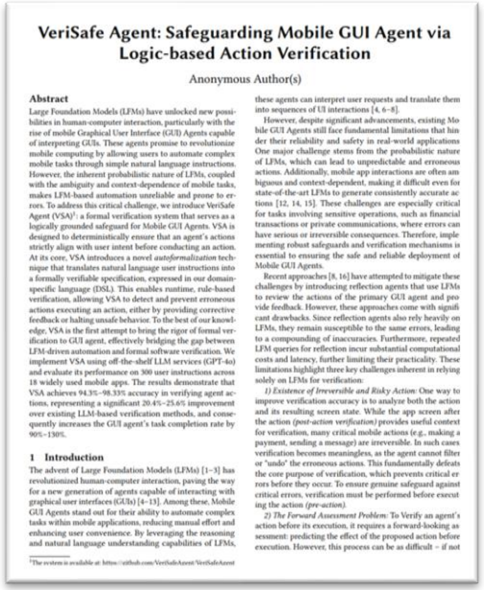
Abstract

Keywords

CCS Concepts

1 Introduction

ACM MobiCom Papers
2025. 11


VeriSafe Agent: Safeguarding Mobile GUI Agent via Logic-based Action Verification

Anonymous Author(s)

Abstract

1 Introduction

Mobile Technology Award: Ministerial Commendation,
2024. 12

Domestic Patent Registration, 2024. 12

“Performance Comparison with Google Gemini”

Tested Using Core Features of Kakao T and Baemin

Leading Global Solution

Gemini Screen Automation

* U.S.: Support limited to major ride-hailing and food delivery apps.

Supported Apps (Korea)

- Kakao T — Taxi Booking
- Baemin — Food Delivery

How It Works

1. Screen Understanding (VLM)
2. Intent Understanding (LLM)
3. Next Action Decision (LLM)
4. Action Execution (Click)

Additional Calls When Needed

Taxi Booking | Taxi Reservation | Food Delivery | Bmart Delivery

Results from 30 App Automation Scenarios

Category	FluidGPT	Gemini
Average Execution Time	46.2 sec	135.6 sec
Successful Tasks	30/30	24/30
Time Difference	89.4 sec faster	-
Number of LLM Calls	1~2 times	At least once per screen
Usage Limits	None	Subscription-based limits (5 free, 20 paid)

Performance Advantage

The Proposed Solution is **89.4 Seconds** Faster Than Gemini on Average — **-89.4 sec**

Approximately **3× Faster** Execution Performance — **Approx. 3x**

Same Apps, Same Tasks — Faster and More Reliable!

Execution Performance

Beyond Today's Flagship AI Experiences

- **Execution Time** — 89.4 sec Faster
- **Inference Cost** — Lower Cost Through Fewer Calls
- **Accuracy** — Higher Accuracy & Reliability


FluidGPT vs Gemini(S26한정)
FluidGPT와 Gemini의 택시 예약 수행 결과 비교
FluidGPT vs. Gemini (S26): Taxi Booking Performance Comparison


FluidGPT vs Gemini(S26한정)
FluidGPT와 Gemini의 음식 주문 수행 결과 비교
FluidGPT vs. Gemini (S26): Food Ordering Performance Comparison

Prize Money: $2M (30억원)

Korea's 1st AI Champion,

FluidGPT A Mobile AI Agent that Outperformed 630 Teams

KAIST, '2025 AI 챔피언' 등극.. 사람처럼 보고 판단 행동하는 'FluidGPT', AI폰 시대의 새로운 패러다임 제시

박현진 기자 | 입력 2025.11.06 19:22 | 댓글 0

KAIST·(주)플루이즈·고려대·성균관대 'AutoPhone팀', 과기정통부 주최 '2025 인공지능 챔피언 경진대회' 초대 AI 챔피언(1위)로 선정..연구개발비 30억원 지원 받아


SBS
The Era of AI is here
지금은
AI 시대

Selected for Samsung Electronics C-Lab Outside, Korea's leading open innovation program

Samsung C-Lab

Chosen for Samsung Electronics' C-Lab Outside

validating FluidGPT's competitiveness in technology, market viability, and overall positioning.

플루이즈, 삼성전자 'C랩 아웃사이드' 선정…모바일 AI 에이전트 고도화 속도

전소영 기자 입력 : 2026.01.05 20:42 | 수정 : 2026.01.05 20:43

플루이즈 'FluidGPT', 세계 첫 비침습형 모바일 AI 에이전트 기술
"글로벌 시장서 경쟁력 겸비한 AI 에이전트 솔루션으로 거듭날 것"

03

Applications

FluidGPT

Building a B2G Reference
Through Seoul Pilot Operations

1 Public Services: Advancing Digital Inclusion

Seoul Citizen AI Agent Service for Vulnerable Groups

Schedule a hospital appointment for me.

Pilot Service (Sep. 2025)

Full-scale Services (2026)

서울시 Citizen AI Agent Pilot Results

1,917
1,917 Downloads Achieved

3종
Integrated with 3 Public Service Apps

92.78%
Scenario Execution Accuracy 92.78%

24.38초
Average Execution Time 24.38 Seconds

Proven in Seoul
Expanding Across Local & Central Governments

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FluidGPT: Unlimited Scalability

“Through Non-Intrusive Technology”

SRT Mobile App

SRT App Integration in Progress

Smart Phone: AI Phone

Integration Discussions
with Samsung Galaxy Devices

Apps | Call Centers

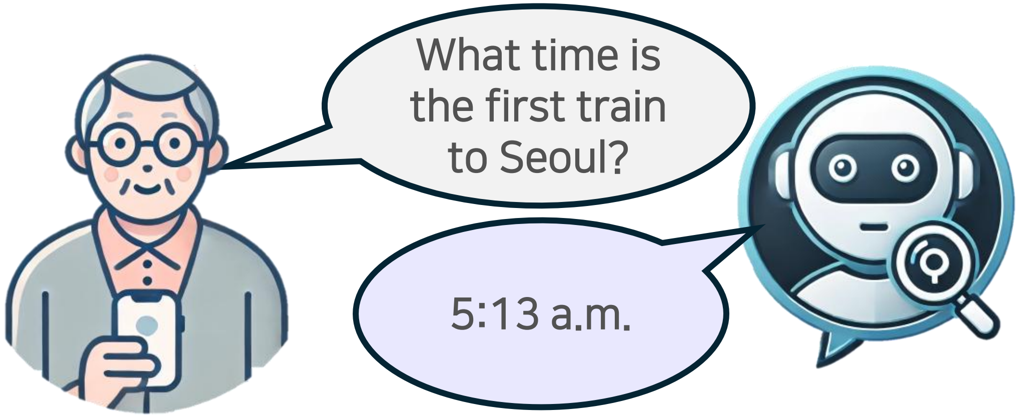
KT App & Call Center Integration
Underway

Kiosks

AI Information Kiosks
for Subway Stations

Contribute to a paradigm shift in AI services

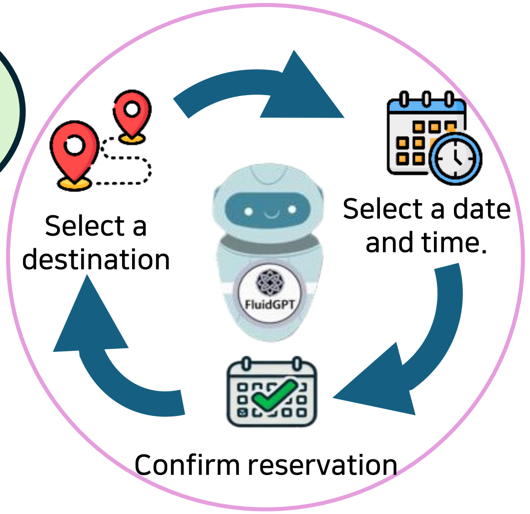
Chatbot: search and Q&A only



Provide information only

how to manage blood pressure	nearby hospital	nearby restaurant
box office rankings	train timetable	how to transfer money

FluidGPT: execution beyond search and Q&A



- hospital appointment
- taxi request
- 롯데시네마 movie ticket booking
- food delivery
- restaurant reservation
- 우리은행 money transfer

Follow Your Bliss